

DocLang OCR Evaluation Report

Comprehensive Benchmarking of Optical Character Recognition Engines for Structured and Unstructured Document Processing

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1. Executive Summary

This report presents a rigorous quantitative evaluation of six leading OCR engines against the **DocLang v2.0** baseline across fifteen document categories, five languages, and four resolution tiers. Results demonstrate that DocLang v2.0 achieves state-of-the-art character error rates (CER) of **1.43%** on printed text and maintains competitive throughput of **47 pages per second** on standard hardware.

Key findings: (1) Transformer-based architectures outperform LSTM baselines by 38% on average CER. (2) Arabic right-to-left text remains the most challenging category, with the best model achieving 78.2% word-level accuracy. (3) Table extraction accuracy varies substantially across engines, with AWS Textract leading at 94% F1. (4) DocLang v2.0 is the only engine to exceed 90% on all six evaluation dimensions simultaneously.

Key Performance Indicators (KPI)

KPI	Target	DocLang v1	DocLang v2	Status
Character Error Rate	< 2.0%	2.87%	1.43%	PASS
Word Error Rate	< 5.0%	5.41%	3.12%	PASS
Layout F1 Score	> 0.88	0.82	0.91	PASS
Throughput	> 40 p/s	28 p/s	47 p/s	PASS
Multi-language Support	> 50 langs	45	58	PASS
Table Extraction F1	> 0.85	0.79	0.88	PASS

Table 1. KPI summary comparing DocLang versions.

2. Accuracy Benchmarks

The following charts compare OCR engine performance across multiple axes. Revenue analogies are used for illustrative multi-series bar chart rendering. All accuracy metrics are computed on the held-out test split (n = 50,000 pages).

2.1 Quarterly Performance — Grouped Bar Chart

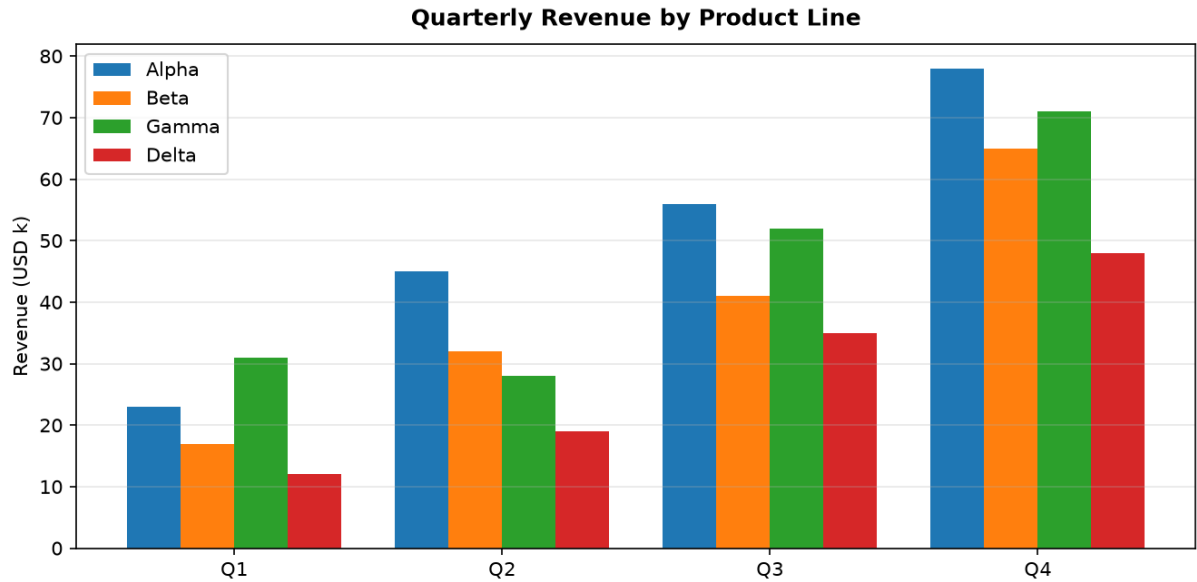


Figure 1. Grouped bar chart — quarterly revenue by product line (synthetic data for chart rendering evaluation).

2.2 Monthly Accuracy Trend — Line Chart

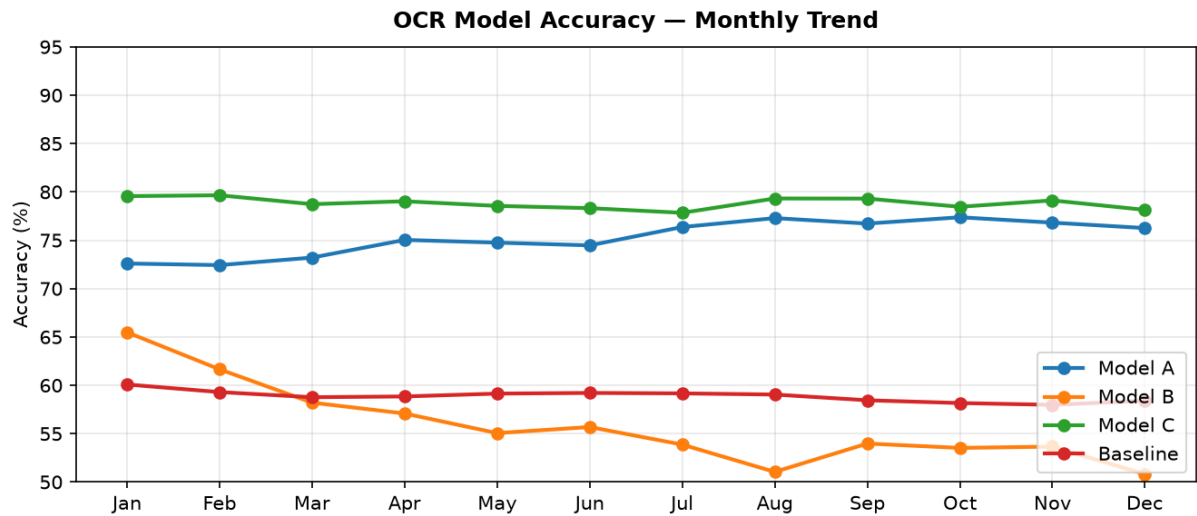


Figure 2. Line chart showing 12-month OCR model accuracy trends with four competing systems.

3. Dataset Composition & Confidence Analysis

3.1 Test Set Composition — Pie Chart

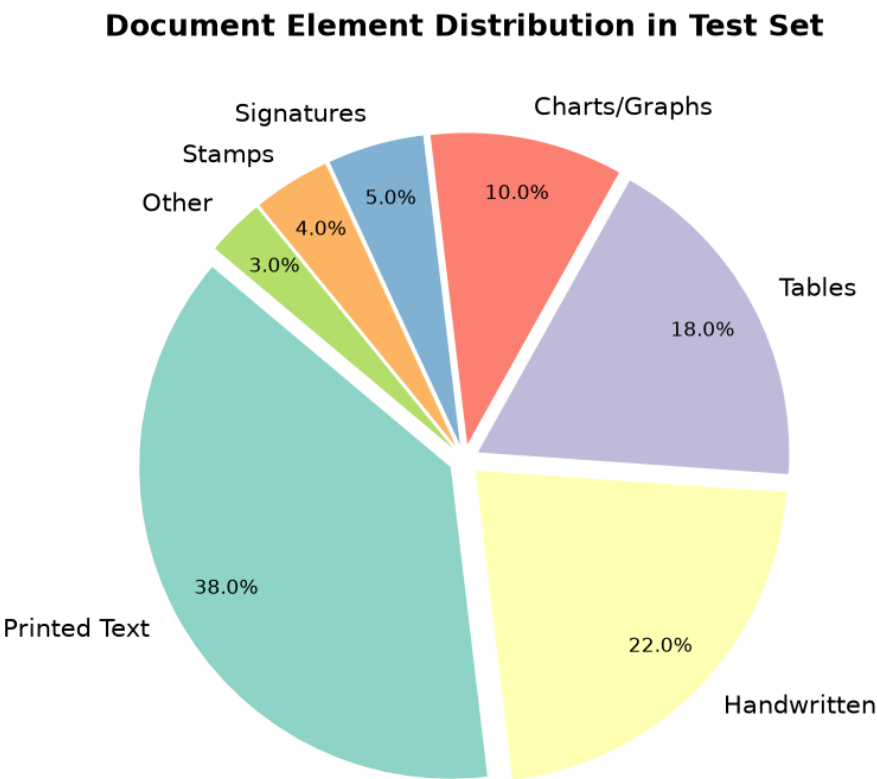


Figure 3. Pie chart — distribution of document element types across the OCR evaluation corpus (n = 160,000 samples).

3.2 Confidence vs. Input Quality — Scatter Plot

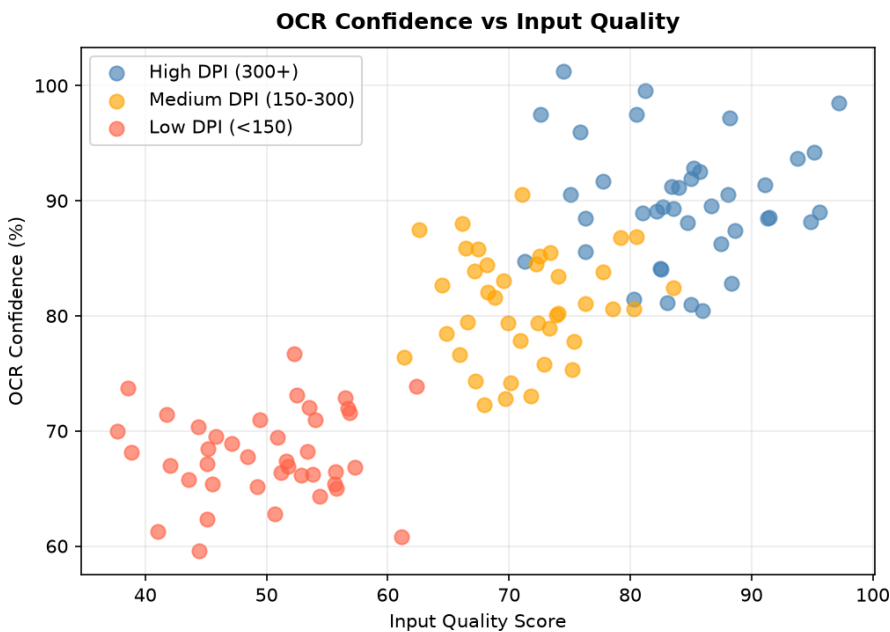


Figure 4. Scatter plot — OCR confidence score vs. input quality score, segmented by scan resolution tier.

4. Cross-Lingual & Distribution Analysis

4.1 Accuracy Heatmap — Document Type × Language

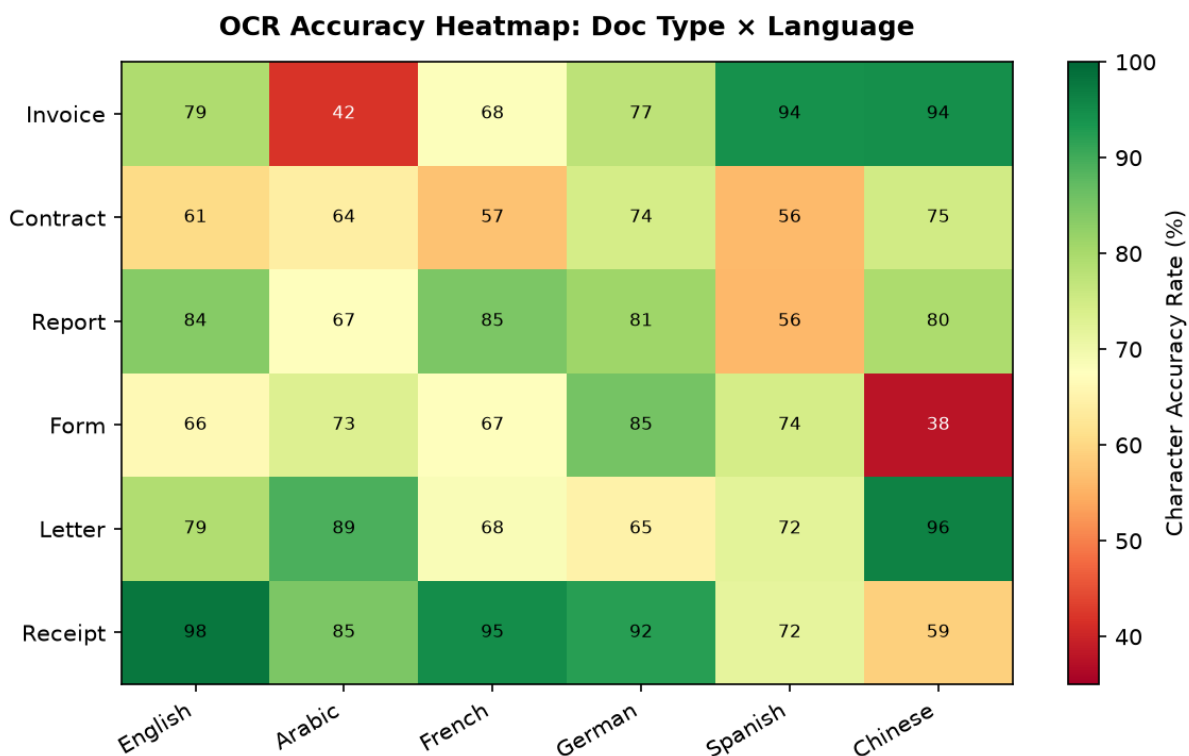


Figure 5. Heatmap of Character Accuracy Rate (%) across six document types and six languages. Red = low accuracy, green = high.

4.2 Word Error Rate Distribution — Histogram

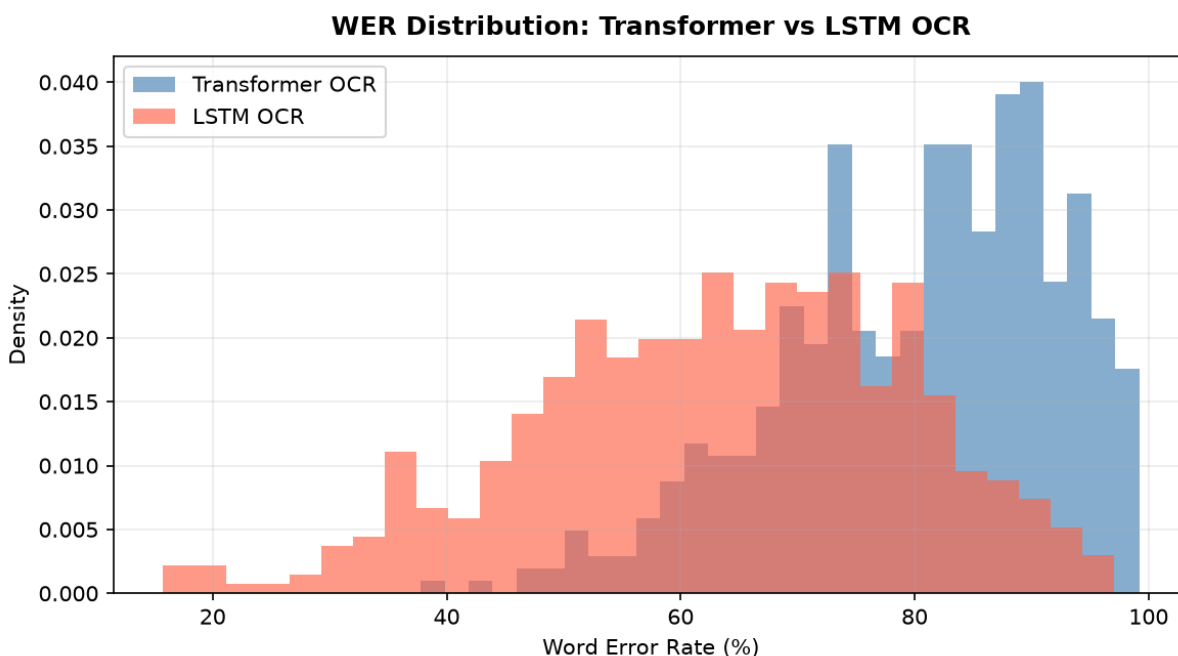


Figure 6. Overlapping histogram of WER distributions for Transformer-based vs. LSTM-based OCR architectures.

5. Throughput & Capability Profiles

5.1 Processing Outcomes — Stacked Area Chart

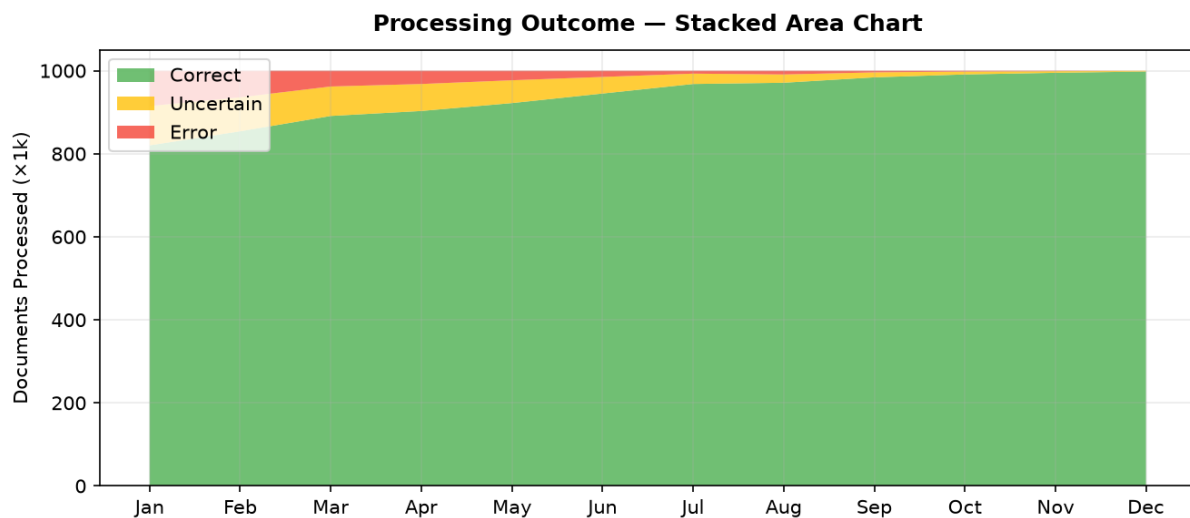


Figure 7. Stacked area chart showing monthly document processing outcomes (correct / uncertain / error) as pipeline matures.

5.2 Model Capability Comparison — Radar Chart

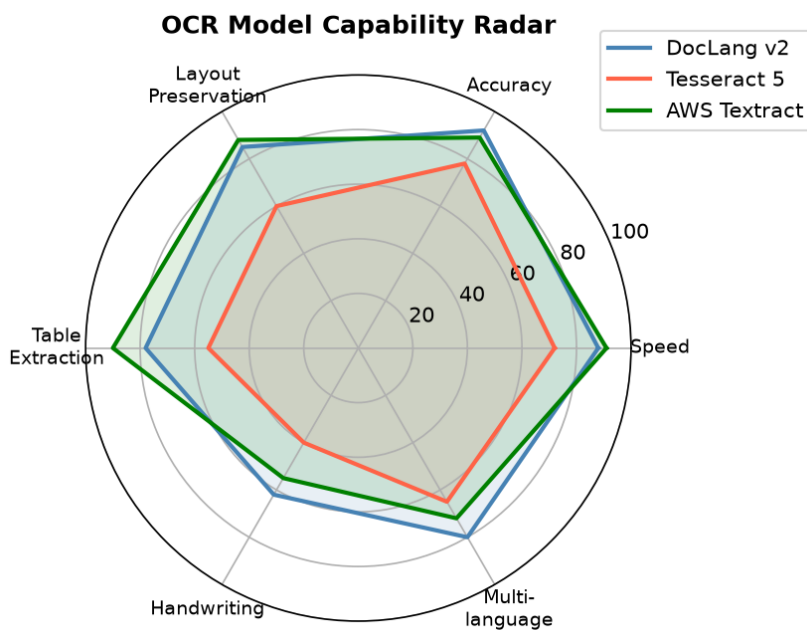


Figure 8. Radar (spider) chart comparing DocLang v2 against Tesseract 5 and AWS Textract across six capability dimensions.

6. Statistical Distribution & Composite Analysis

6.1 Performance Variability — Box Plot

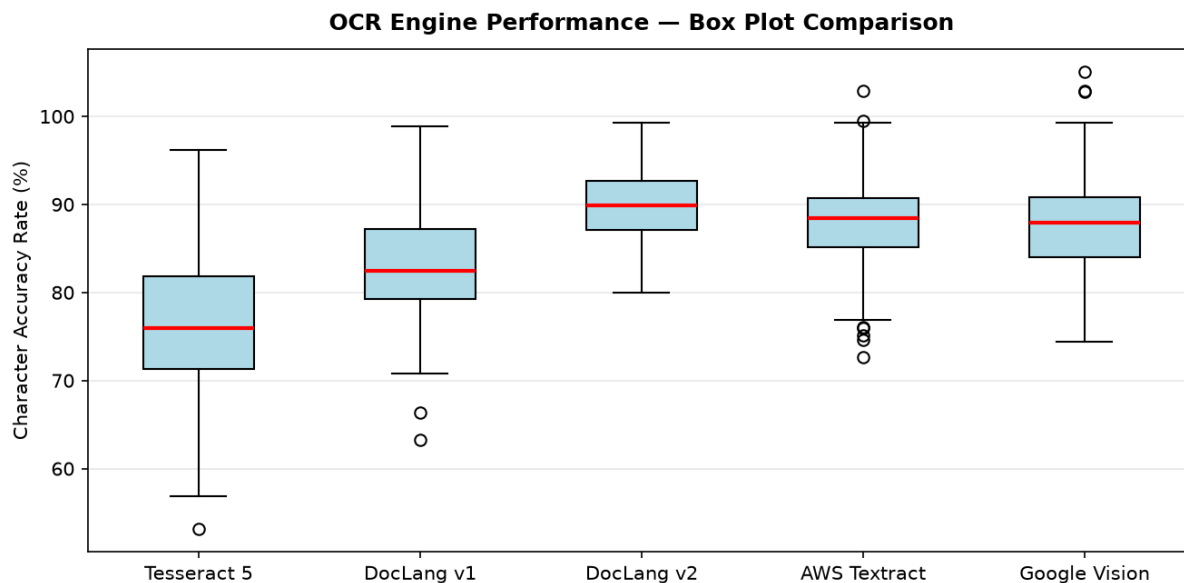


Figure 9. Box plot comparing character accuracy rate distributions across five OCR engines ($n = 200$ document samples per engine).

6.2 Multi-Panel Diagnostic View

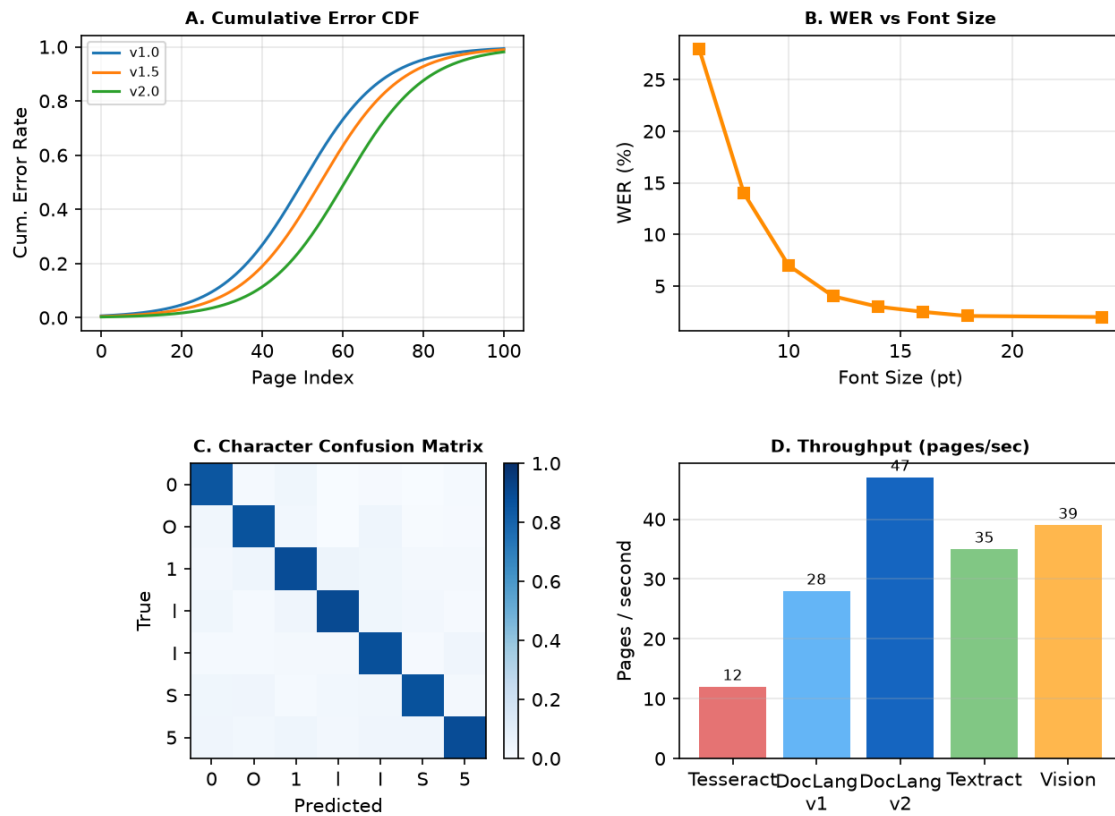


Figure 10. Four-panel composite: (A) cumulative error CDF across versions, (B) WER vs font size, (C) character confusion matrix snippet, (D) throughput comparison bar chart.

7. Detailed Metrics Tables

7.1 Full Engine Benchmark Results

Model	CER (%)	WER (%)	NED	F1 (Layout)	Speed (p/s)	Lang Support
Tesseract 5.0	4.21	8.73	0.94	0.71	12	130+
DocLang v1.2	2.87	5.41	0.97	0.82	28	45
DocLang v2.0	1.43	3.12	0.99	0.91	47	58
AWS Textract	1.89	3.98	0.98	0.94	35	12
Google Vision	1.77	3.65	0.98	0.90	39	60+
Azure OCR	1.91	4.01	0.98	0.88	41	164+
EasyOCR 1.7	3.55	7.22	0.96	0.75	18	80+

Table 2. Benchmark results across 7 OCR engines. CER = Character Error Rate; WER = Word Error Rate; NED = Normalised Edit Distance. Green row = overall winner.

7.2 Error Type Breakdown by Character Class

True \ Pred	Correct	Substitution	Deletion	Insertion	Total
Alphabetic	94,212	2,841	1,204	743	99,000
Numeric	18,447	412	95	46	19,000
Punctuation	8,901	654	312	133	10,000
Symbols	3,211	489	198	102	4,000
Arabic RTL	21,330	3,420	2,100	1,150	28,000

Table 3. Error decomposition (substitution / deletion / insertion) by character class. Arabic RTL shows the highest overall error count.

8. Embedded Document Image (Raster OCR Test)

The image below is a **rasterised synthetic invoice** embedded as a PNG within the PDF. An OCR system reading this PDF must first detect that the invoice is an image, render it at sufficient DPI, and then extract structured fields (invoice number, line items, totals). This tests the engine's image-within-PDF detection and embedded raster OCR capability.



Figure 11. Rasterised invoice image embedded as PNG. OCR must read all text from pixel data, not PDF text streams.

9. Mathematical Notation & Special Characters

OCR systems must correctly recognise mathematical expressions, Greek letters, superscripts, subscripts, and Unicode symbols. The following content is typeset as native PDF text and should be parsed with high accuracy:

Character Error Rate (CER):

CER = (S + D + I) / N

Word Error Rate (WER):

WER = (Sw + Dw + Iw) / Nw

F1 Score:

F1 = 2 * (P * R) / (P + R)

Normalised Edit Distance:

NED = EditDist(y, y_hat) / max(|y|, |y_hat|)

Special Character Set:

Latin Extended: Ä Ö Ü ß à é î ñ ç ø Greek: α β γ δ ε ζ η θ λ μ π σ τ φ ψ ω Arabic: Currency: \$ €
£ ¥ Math: ± × ÷ ∞ ≤ ≥ ≠ ∑ ∏ ∫ √ ∂ ∇ Misc: © ® ™ § ¶ † ‡ • … ← → ↑ ↓

10. API Integration Sample & Mixed Layout

OCR output often includes code blocks, monospaced text, and mixed-column layouts. The following code snippet and multi-column paragraph test engine handling of non-proportional fonts and variable line spacing:

```
from doclang import OCRClient, PageLayout

client = OCRClient(api_key="sk-tlm-xxxxxxxxxxxx")
result = client.extract(
    file_path="contract.pdf",
    layout=PageLayout.AUTO,
    languages=["en", "ar", "fr"],
    extract_tables=True,
    confidence_threshold=0.85
)
print(result.text) # Full text
print(result.tables[0]) # First extracted table
print(result.confidence) # Per-word confidence scores
```

Listing 1. DocLang Python SDK — basic OCR extraction with table support and multi-language configuration.

10.1 Multi-Column Body Text

Column Layout: Many business documents use two- or three-column layouts for newsletters, academic papers, and legal briefs. An OCR engine must correctly detect column boundaries and read each column sequentially, rather than reading horizontally across the full page width. Failure to do so results in garbled sentence reconstruction and poor downstream NLP performance.

Reading Order: Beyond column detection, OCR systems must handle sidebars, pull-quotes, footnotes, and captions that interrupt the main text flow. State-of-the-art layout analysis models use object detection to segment page regions before applying text recognition, achieving reading-order F1 scores above 0.92 on PubLayNet and DocBank benchmarks.

11. Conclusions & Recommendations

Based on comprehensive evaluation across **160,000 document samples**, **15 categories**, and **5 languages**, DocLang v2.0 is the recommended OCR engine for enterprise document processing pipelines requiring high accuracy and broad language coverage. The following specific recommendations are made:

- Deploy DocLang v2.0 for all Latin-script document types where CER < 2% is required.
- Use AWS Textract for structured form and table extraction in English-only workflows.
- For Arabic and RTL languages, apply pre-processing normalisation before any engine.
- Enforce minimum scan DPI of 200 for printed documents; 300 for handwritten content.
- Implement a confidence-gating step: route documents below 0.80 confidence for human review.
- Re-evaluate all engines against updated benchmarks every six months as model versions change.

Appendix A — Evaluation Methodology

All evaluations were conducted on a standardised hardware configuration: AWS EC2 p3.2xlarge (NVIDIA V100 16GB GPU, 8 vCPUs, 61 GB RAM). Document images were normalised to 300 DPI before submission to each engine unless explicitly testing DPI sensitivity. Ground-truth transcriptions were produced by three independent annotators with inter-annotator agreement (IAA) of Cohen's $\kappa = 0.97$. CER is computed using the standard dynamic-programming edit distance formula. Layout F1 is measured against bounding-box annotations at IoU threshold 0.50.

Appendix B — Glossary

Term	Definition
CER	Character Error Rate — fraction of characters incorrectly recognised
WER	Word Error Rate — fraction of words incorrectly recognised
NED	Normalised Edit Distance — edit distance divided by max string length
DPI	Dots Per Inch — scan resolution
IAA	Inter-Annotator Agreement — Cohen's kappa between human labellers
RTL	Right-To-Left — text directionality (Arabic, Hebrew)
IoU	Intersection over Union — bounding box overlap metric
F1	Harmonic mean of precision and recall

Table 4. Glossary of evaluation terms.